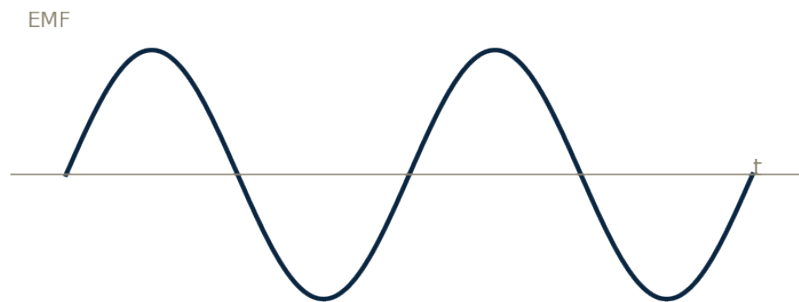


Electromagnetic Induction & AC

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Faraday's Law of Electromagnetic Induction: the induced EMF equals the negative rate of change of magnetic flux, $\epsilon = -d(\phi)/dt$.
- Lenz's Law: the induced current opposes the change in magnetic flux that produced it (conservation of energy).
- Motional EMF: for a rod of length L moving with velocity v perpendicular to a magnetic field B , $\epsilon = BLv$.
- Self-inductance L relates the induced EMF in a coil to the rate of change of current in itself: $\epsilon = -L(di/dt)$. Mutual inductance M relates EMF in one coil to changing current in another.
- In an AC circuit, the current and voltage are represented as $I = I_0 \sin(\omega t)$ and $V = V_0 \sin(\omega t + \phi)$, where ϕ is the phase difference (depends on the circuit elements).
- For a pure resistor, V and I are in phase. For a pure inductor, current lags voltage by 90 degrees. For a pure capacitor, current leads voltage by 90 degrees.
- RMS value of AC: $I_{\text{rms}} = I_0/\sqrt{2}$; this is the value used for power calculations ($P = I_{\text{rms}}^2 R$).
- In an LCR series circuit, impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$, where $X_L = \omega L$ and $X_C = 1/(\omega C)$. Resonance occurs when $X_L = X_C$.



A sinusoidal AC waveform showing EMF varying with time.

2. Formula Sheet

Faraday's law	$\epsilon = -d(\phi)/dt$
Motional EMF	$\epsilon = BLv$
Self-induced EMF	$\epsilon = -L (dI/dt)$
RMS value of AC current	$I_{rms} = I_0/\sqrt{2}$
Inductive reactance	$X_L = \omega L$
Capacitive reactance	$X_C = 1/(\omega C)$
Impedance of LCR circuit	$Z = \sqrt{R^2 + (X_L - X_C)^2}$
Resonant frequency of LCR circuit	$f_0 = 1 / (2 \pi \sqrt{LC})$

3. Practice Problems (22 questions)

1. A coil has a magnetic flux of 0.02 Wb passing through it. If the flux drops to zero in 0.1 s, find the average induced EMF.
2. A rod of length 0.5 m moves with a velocity of 4 m/s perpendicular to a magnetic field of 0.2 T. Find the motional EMF induced.
3. State Lenz's law and explain how it is consistent with the law of conservation of energy.
4. A coil of self-inductance 2 H carries a current that changes at the rate of 3 A/s. Find the induced EMF.
5. An AC voltage has a peak value of 220 V. Find its RMS value.
6. Find the inductive reactance of a 0.5 H inductor at a frequency of 50 Hz ($\omega = 2 \pi f$).
7. Find the capacitive reactance of a 10 microF capacitor at a frequency of 50 Hz.
8. In an LCR series circuit, $R = 30 \text{ ohm}$, $X_L = 50 \text{ ohm}$, $X_C = 10 \text{ ohm}$. Find the impedance of the circuit.
9. Explain, using Lenz's law, why a metal ring dropped through a strong magnetic field falls more slowly than in the absence of the field (eddy current damping).
10. Find the resonant frequency of an LCR circuit with $L = 1 \text{ H}$ and $C = 1 \text{ microF}$.
11. A generator coil of 100 turns and area 0.1 m^2 rotates at 50 rev/s in a magnetic field of 0.2 T. Find the peak EMF generated.
12. State the phase relationship between current and voltage in (a) a pure resistor (b) a pure inductor (c) a pure capacitor.
13. A transformer has 200 turns in the primary and 1000 turns in the secondary. If the primary voltage is 220 V, find the secondary voltage (assume an ideal transformer).
14. Explain why AC is preferred over DC for long-distance power transmission.
15. Two coils have a mutual inductance of 0.5 H. If the current in the primary coil changes at 4 A/s, find the EMF induced in the secondary coil.

16. In an AC circuit with only an inductor connected to a 100 V (RMS) supply of frequency 50 Hz, and $L = 0.2 \text{ H}$, find the RMS current.
17. Explain the working principle of an AC generator based on Faraday's law of electromagnetic induction.
18. A step-down transformer converts 440 V to 220 V. If the primary has 2000 turns, find the number of turns in the secondary.
19. Find the average power dissipated in a pure inductor connected to an AC source, and explain why it is zero.
20. At resonance in a series LCR circuit, what is the value of the impedance, and how does the current compare to its value at other frequencies?
21. A coil of 500 turns experiences a change in flux from 0.01 Wb to 0.05 Wb in 0.2 s. Find the average induced EMF.
22. Explain, using Faraday's law, why no current is induced in a coil kept stationary in a uniform, unchanging magnetic field.

Answer Key

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|---|---|
| 1. $e = 0.2 \text{ V}$ | 2. $e = 0.4 \text{ V}$ |
| 3. Conceptual - opposing induced current requires work to be done, consistent with energy conservation (no free energy) | 4. $e = 6 \text{ V}$ |
| 5. $V(\text{rms})$ approx 155.6 V | 6. $X_L = 157 \text{ ohm}$ (approx) |
| 7. $X_C = 318 \text{ ohm}$ (approx) | 8. $Z = 50 \text{ ohm}$ |
| 9. Conceptual - eddy currents induced in the ring oppose its motion (Lenz's law), creating a retarding force | 10. f_0 approx 159 Hz |
| 11. $\text{EMF}(\text{peak}) = 2\pi \times 50 \times 100 \times 0.1 \times 0.2 = 628 \text{ V}$ (approx) | 12. Conceptual - (a) in phase (b) current lags voltage by 90 degrees (c) current leads voltage by 90 degrees |
| 13. $V(\text{secondary}) = 1100 \text{ V}$ | 14. Conceptual - AC can be easily stepped up/down using transformers, reducing transmission (I^2R) losses at high voltage |
| 15. $e = 2 \text{ V}$ | 16. $I(\text{rms})$ approx 1.59 A |
| 17. Conceptual - rotating coil in a magnetic field changes flux, inducing an alternating EMF (Faraday's law) | 18. $N(\text{secondary}) = 1000 \text{ turns}$ |
| 19. Conceptual - average power is zero since current and voltage are 90 degrees out of phase (no net energy transfer per cycle) | 20. Conceptual - impedance is minimum ($Z=R$) at resonance; current is maximum at resonance |
| 21. $e = 1 \text{ V}$ | 22. Conceptual - flux is constant, so $d(\phi)/dt = 0$, giving zero induced EMF (Faraday's law) |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.